The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.4.4 (prototype)
Log entry date	14 October 2026
Status	Released: the code was pushed to the GitHub main branch in commit 7ba922c; the 0.4.4 documents and this log follow in a separate commit. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch (commit 7ba922c, Release 0.4.4)
Previous version	0.4.3, 10 October 2026, commit fa9ae51 (see Spaghetti_Engine_Dev_Log_v0.4.3.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.4.3 (10 October 2026) to version 0.4.4 of The Spaghetti Engine. Version 0.4.4 changes the Transition Pathways tab: an outcome weight or flat-response threshold left blank in the setup now takes its default value instead of 0, the analysis accepts expert-defined sequences, and the robustness test covers outcome weights and archetype thresholds as well as simulation settings.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.4.4

Area	Change
Blank inputs	An outcome weight or flat-response threshold left blank in the setup now falls back to its default (weight 1, threshold 0.015) instead of being read as 0.
All weights 0	The analysis can no longer be run when every outcome has weight 0; a note explains why.
Expert-defined sequences	Ordered stages, each at the conditional transition point or at a fixed intensity, simulated and classified together with the enumerated pathways.
Robustness	Three families under the existing switch: simulation settings, outcome weights, and archetype thresholds, each with its own shares.
Regret	The tab and the workbook state that the regret is taken over converged pathways only.
About page	States when the documentation describes an earlier version than the running tool.
Record of 0.4.3	Version 0.4.3 was pushed as commit fa9ae51 on 10 October 2026 and deployed the same day.

3. The Changes in Detail

3.1 Blank weights and thresholds

- **Default values.** An outcome weight or flat-response threshold left blank in the Transition Pathways setup now falls back to its default, a weight of 1 and a threshold of 0.015. Before, a cleared field was read as 0: a blank weight silently removed that outcome from the total outcome gain, and a blank threshold set the flat-response threshold to 0.
- **All weights 0.** When every outcome has weight 0, the total outcome gain is 0 for every pathway and no ranking means anything. The analysis can then no longer be run, and a note explains why.

3.2 Expert-defined sequences

- **Sequences.** An expert-defined sequence is an ordered list of stages, each at the conditional transition point or at a fixed intensity on the sweep grid. It is simulated with the same engine as the enumerated pathways, flagged in the results table, the CSV export, and the workbook, and classified with the enumerated pathways.
- **Thresholds.** Expert-defined sequences count toward the quantile thresholds of the archetypes, because the methods evaluate those thresholds over all admissible candidates.

3.3 Robustness and regret

- **Three families.** The robustness switch now runs three families: simulation settings (as before), outcome weights (each weight doubled with the others at 1, with a full re-run), and archetype thresholds (effort quartiles 0.20/0.80 and 0.30/0.70, top share 0.75 and 0.85, average transition point cut-offs one grid step lower and higher, and the EBI cut-off at 0.45 and 0.55; re-classification only). The tab and the workbook report the shares per family.
- **Regret.** The best total outcome gain and the regret are taken over converged pathways only, as before; the tab and the workbook now say so.

3.4 About page

- **Documentation note.** The About page states when the documentation describes an earlier version than the running tool, instead of always saying that it describes the version in use.

4. Implementation Choices

Topic	Choice in the tool	Reason
Blank weight or threshold	The default value (weight 1, threshold 0.015).	A cleared number field holds an empty string, which reads as 0; a field left empty is far more likely an unfinished edit than a deliberate 0, which can still be typed.
Every weight 0	The analysis cannot be run; a note says why.	Every pathway would have a total outcome gain of 0, so rankings and archetypes would be meaningless.
Expert sequences and thresholds	Counted toward the quantile thresholds.	The methods evaluate the thresholds over all admissible candidates.
Threshold family	Re-classification only.	The archetype thresholds do not change the simulations, so no re-run is needed.

5. Effect on Existing Results

- Results of 0.4.3 are unchanged, unless an outcome weight or the flat-response threshold was left blank in the Transition Pathways setup: such an analysis used 0 and should be re-run.
- Without expert-defined sequences, the enumerated pathways and their archetypes are computed as in 0.4.3.

- With robustness switched on, the analysis also runs the outcome-weight family (a full re-run per outcome) and takes longer than in 0.4.3.
- Version raised from 0.4.3 to 0.4.4 in package.json and package-lock.json.

6. Defects Found and Fixed

Problem	Cause	Fix
A blank outcome weight removed that outcome from the total outcome gain, without a warning.	A cleared number field holds an empty string, and the number of an empty string is 0.	A blank weight falls back to 1.
A blank flat-response threshold set the threshold to 0.	The same.	A blank threshold falls back to 0.015.

7. Verification

- **Tests:** tests/test_pathway_blank_inputs.cjs (new, 5 checks): a blank or non-numeric field takes its default, an explicit 0 is kept, and a blank weight or threshold gives the same result as its default. tests/test_pathway_expert_robustness.cjs (new, 28 checks): expert-defined sequences and the three robustness families. tests/test_marginal_benefit_tp.cjs (37 checks) and tests/test_transition_direction.cjs (37 checks) still pass.
- **Build:** the production build completes without errors.

8. Deployment Status

- **Deployed** with the push of release commit 7ba922c to main on 14 October 2026 (pull requests #1 and #2). Netlify rebuilds the site automatically. The 0.4.4 documents and this log follow in a separate commit.
- **Files changed:** src/SpaghettiEngine.jsx, src/AboutTab.jsx, tests/test_pathway_blank_inputs.cjs (new), tests/test_pathway_expert_robustness.cjs (new), package.json and package-lock.json, and README.md; with the documents, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.4.4 versions replace those of 0.4.3.
- **Build note:** npm run build fails in the local OneDrive folder because npm's command shim cannot handle the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.
- **PDF note:** as for 0.4.3, the PDFs were printed with Chrome (EDGE_PATH set to chrome.exe), because Edge's headless printing produced no file on the build machine.

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.4.3 logs still apply.
- The User Manual and the Methodological Annex of version 0.4.4 do not yet describe the expert-defined sequences and the three robustness families; until they do, Sections 3.2 and 3.3 of this log describe them.
- The regression tests of 0.4.0 are still not in the repository; tests/ holds the four test files named above.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.4.4)	How to use every feature. Section 1.2 lists the changes of 0.4.3; Sections 12 and 13 describe the Transition Point and Transition Pathways tabs.
The Spaghetti Engine - Methodological Annex (version 0.4.4)	The methods. Section 12.1 defines the marginal-benefit transition point, Section 13 the transition pathway analysis, and Section 13.11 the robustness to simulation settings.
Spaghetti_Engine_Dev_Log_v0.4.3	The 0.4.3 release: the marginal-benefit transition point.
Spaghetti_Engine_Dev_Log_v0.4.2	The 0.4.2 release: readable text and the Change at TP column.
This Development Log	What changed in 0.4.4 and the technical work behind it.

11. Next Steps

- Describe the expert-defined sequences and the three robustness families in the User Manual and the Methodological Annex.
- Decide whether a stage should look past an early dip in the marginal benefit when it rises again later (Section 9 of the 0.4.3 log).
- Rebuild the full engine regression set inside tests/.
- Continue with the development roadmap: version 0.5.0 (guided first use) is next.